
Transfer Learning and Open-Set Recognition for Multi-Label Image Classification

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Abstract

1 Introduction

Deep neural networks have achieved strong performance in image classification tasks, but these successes typically rely on large labeled datasets and closed-set assumptions in which all test-time categories are known during training. In practical settings, however, labeled data may be limited and models may encounter novel classes at inference time, making standard classification pipelines unreliable. These challenges motivate approaches that are able to utilize both pretrained representations and explicitly handle open-set uncertainty.

Transfer learning addresses data scarcity by adapting models pretrained on large-scale datasets such as ImageNet to downstream tasks. While this strategy improves closed-set performance, it does not inherently enable models to detect unseen classes. Open-set recognition (OSR) extends the classification capabilities by requiring models to identify out-of-distribution inputs rather than forcing confident but incorrect predictions. Designing reliable OSR methods is particularly challenging in fine-grained settings, where unseen classes may closely resemble known ones.

In this project, we study transfer learning and multiple open-set recognition methods in a hierarchical image classification task involving three superclasses (bird, dog, reptile) and 87 fine-grained subclasses. Using a provided training dataset, we evaluate several ImageNet-pretrained backbones under a unified training protocol and compare confidence-based and prototype-based OSR methods. Our experiments show that confidence-based OSR provides robust and consistent performance across architectures, while prototype-based cosine distance methods fail due to misalignment between embedding geometry and class structure.

2 Related Work

Transfer learning using ImageNet-pretrained models has become a standard approach for visual recognition under limited labeled data, with architectures such as ResNet, DenseNet, EfficientNet, and Vision Transformers demonstrating strong cross-domain generalization (He et al. [2016], Huang et al. [2017], Tan and Le [2019]). Open-set recognition extends closed-set classification by requiring models to detect unseen categories at inference time. Prior approaches include confidence-based methods, activation distribution modeling (e.g. OpenMax), distance-based techniques using Mahalanobis or cosine metrics, and energy-based scoring (Lee et al. [2018], Liu et al. [2020]). Metric-learning methods such as prototypical networks represent classes via embedding-space centroids, but their effectiveness depends on the geometry of the learned representation (Snell et al. [2017]). Recently, vision–language models such as CLIP have demonstrated improved robustness under distribution shift, motivating their use in open-set and OOD settings (Radford et al. [2021], Ming et al. [2022]).

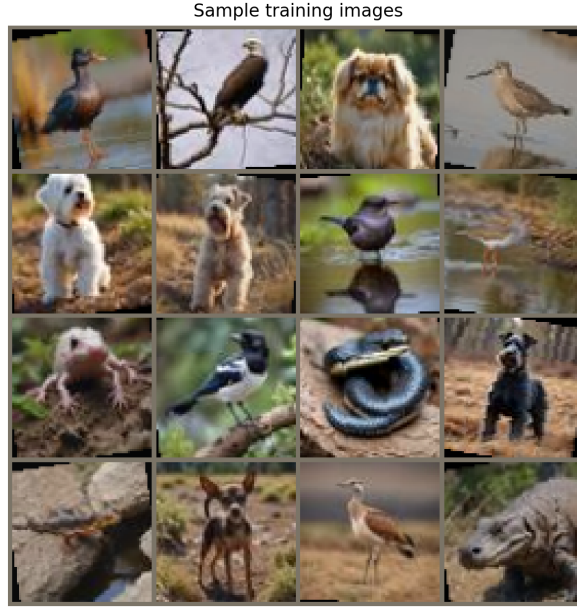


Figure 1: Example training images after augmentation.

3 Dataset and Preprocessing

3.1 Dataset Description and Split

The provided dataset consists of 6,600 RGB images at 64×64 resolution, labeled with a coarse superclass (3 categories: bird, dog, reptile) and a fine-grained subclass (87 categories).

We split all labeled training data into:

- **Train split:** 70% of data, stratified by superclass.
- **Validation split:** 15%, used to tune hyperparameters and select the best baseline and transfer learning models.
- **Calibration split:** 15%, used exclusively for threshold selection in Open-Set Recognition (OSR) experiments.

We ensure the splits preserve superclass proportions. Subclass overlap is allowed because OSR methods operate only at inference time.

3.2 Preprocessing Pipeline: Data Augmentation and Normalization

All images are resized to 224×224 pixels and normalized using ImageNet mean and standard deviation to ensure compatibility with ImageNet-pretrained backbones and consistent training across all models.

For training models, we apply a strong but standard data augmentation procedure consisting of: random horizontal flip, random rotation ($\pm 10^\circ$), random resized crop, and color jitter (brightness, contrast, saturation). A grid of examples of augmented images is shown in Figure 1. During training, each epoch presents a different randomly-transformed version of the same images, effectively expanding the number of unique training samples without modifying the dataset itself.

4 Transfer Learning

We employ transfer learning with ImageNet-pretrained visual backbones to address data scarcity and encourage effective feature reuse. Each model comprises a shared feature extractor followed by two

parallel classification heads: (i) a superclass head predicting four categories (bird, dog, reptile, novel), and (ii) a subclass head predicting 88 categories (87 seen subclasses plus a novel class).

4.1 Backbone Choice

EfficientNet, DenseNet, ResNet, and CLIP backbones are compared under a consistent transfer-learning setup to assess the impact of architectural design and pretraining regime. EfficientNet and DenseNet emphasize parameter efficiency and feature reuse through compound scaling and dense connectivity, while ResNet serves as a widely adopted convolutional baseline with residual learning. In contrast, CLIP (ViT-B/32) employs a transformer-based visual encoder pretrained on large-scale image-text data. In our setup, we use only the visual encoder of CLIP. Prior work has shown that, even when used solely as an image encoder, CLIP learns visual representations that generalize better under distribution shift, are less reliant on dataset-specific shortcuts, and transfer effectively to downstream tasks with limited supervision [Radford et al., 2021, Ming et al., 2022].

4.2 Training Strategy

Training is performed in two phases. In the first phase, the pretrained backbone is frozen, and only the two classification heads are optimized using cross-entropy loss for up to 15 epochs. This allows the model to learn task-specific decision boundaries while preserving generic visual representations. In the second phase, the backbone is unfrozen and fine-tuned end-to-end together with the classification heads for up to 10 additional epochs, using a smaller learning rate and a cosine decay schedule to ensure stable adaptation and prevent catastrophic forgetting of pretrained features.

To mitigate overfitting, we incorporate multiple regularization mechanisms. First, we employ early stopping based on validation loss with a patience of 3 epochs; in practice, this criterion was not triggered, as validation performance continued to improve or stabilize throughout training. Second, we use the AdamW optimizer, which decouples weight decay from gradient updates and provides implicit regularization. Finally, during the fine-tuning stage, we apply a cosine learning-rate decay schedule to promote smooth convergence and stable adaptation.

Algorithm 1 summarizes the preprocessing and transfer-learning procedure used for all pretrained backbones.

5 Open-Set Recognition

In addition to closed-set hierarchical classification, we consider open-set recognition (OSR) in which test samples may belong to unseen superclasses or subclasses. The goal is to correctly classify in-distribution (ID) samples while identifying out-of-distribution (OOD) inputs and assigning them to a novel class.

5.1 Local Calibration Set

To enable reliable open-set detection, we construct an augmented calibration set containing both in-distribution (ID) and out-of-distribution (OOD) samples. The ID portion is the held-out calibration dataset. OOD samples consist of near-OOD and far-OOD categories capturing different degrees of distributional shift.

Near-OOD samples are synthesized from ID using a strong augmentation pipeline designed to preserve coarse semantic structure while substantially altering appearance. Specifically, we apply extreme random resized crops with a small retained area, grayscale conversion, Gaussian blur, and aggressive color jitter, followed by ImageNet normalization.

Far-OOD samples are drawn from the external CIFAR-100 dataset [Krizhevsky and Hinton, 2009] and restricted to animal categories. To avoid semantic leakage, we explicitly exclude bird, reptile, and dog classes, ensuring that these samples represent genuinely novel categories with respect to the in-distribution label space.

Algorithm 1 Transfer Learning with Dataset Preprocessing

Require: Dataset $\mathcal{D} = \{(x_i, y_i^{\text{super}}, y_i^{\text{sub}})\}$ with splits $\mathcal{D}_{\text{train}}, \mathcal{D}_{\text{val}}$

Require: Training transform $\mathcal{T}_{\text{train}}$; evaluation transform $\mathcal{T}_{\text{eval}}$

Require: Pretrained backbone f_θ ; classifier heads $g_\phi^{\text{super}}, g_\phi^{\text{sub}}$

Require: Loss weight λ ; epochs E_1 (frozen), E_2 (fine-tune)

Ensure: Trained parameters (θ, ϕ)

```
1: (1) Preprocessing
2: for  $(x, y^{\text{super}}, y^{\text{sub}}) \in \mathcal{D}_{\text{train}}$  do
3:    $x' \leftarrow \mathcal{T}_{\text{train}}(x)$ 
4: end for
5: for  $x \in \mathcal{D}_{\text{val}}$  do
6:    $x' \leftarrow \mathcal{T}_{\text{eval}}(x)$ 
7: end for

8: (2) Train classifier heads with frozen backbone
9: Freeze backbone parameters  $\theta$ 
10: for epoch = 1 to  $E_1$  do
11:   for  $(x', y^{\text{super}}, y^{\text{sub}}) \sim \mathcal{D}_{\text{train}}$  do
12:      $h \leftarrow f_\theta(x')$ 
13:      $\hat{p}^{\text{super}} \leftarrow \text{softmax}(g_\phi^{\text{super}}(h))$ 
14:      $\hat{p}^{\text{sub}} \leftarrow \text{softmax}(g_\phi^{\text{sub}}(h))$ 
15:      $\mathcal{L} \leftarrow \text{CE}(\hat{p}^{\text{super}}, y^{\text{super}}) + \lambda \text{CE}(\hat{p}^{\text{sub}}, y^{\text{sub}})$ 
16:     Update  $\phi$ 
17:   end for
18: end for

19: (3) Fine-tune backbone and heads
20: Unfreeze backbone parameters  $\theta$ 
21: for epoch = 1 to  $E_2$  do
22:   for  $(x', y^{\text{super}}, y^{\text{sub}}) \sim \mathcal{D}_{\text{train}}$  do
23:     Compute  $\mathcal{L}$  as above
24:     Update  $(\theta, \phi)$ 
25:   end for
26: end for
27: return  $(\theta, \phi)$ 
```

5.2 Confidence-Based OSR

Confidence-based OSR is derived from the classifier’s softmax outputs. For each input image, the maximum predicted class probability is used as a confidence score.

In Figure 2, We show the empirical distributions of confidence scores for ID, near-OOD, and far-OOD samples. ID samples concentrate at high confidence values, while both near-OOD and far-OOD samples exhibit lower confidence. Notably, near-OOD samples generated via strong appearance-altering transformations, tend to produce extremely low confidence scores, indicating a pronounced sensitivity of the model to severe local distributional shifts.

Accordingly, we set the decision threshold τ as the midpoint between the lower tail of the ID confidence distribution and the upper tail of the far-OOD confidence distribution, computed as

$$\tau = \frac{1}{2} (\text{ID}_{5\%} + \text{far-OOD}_{95\%}).$$

5.3 Prototype-Based OSR

We also implemented prototype-based cosine distance OSR as a complement to confidence OSR with the intention of combining the two predictions to improve accuracy. Class prototypes were computed by averaging embedding vectors over training samples, and test samples were rejected as novel when their minimum cosine distance to all prototypes exceeded a calibrated threshold. We assumed that the embedding space would be suitable for prototype OSR and assumed a starting threshold of 0.95/0.05.

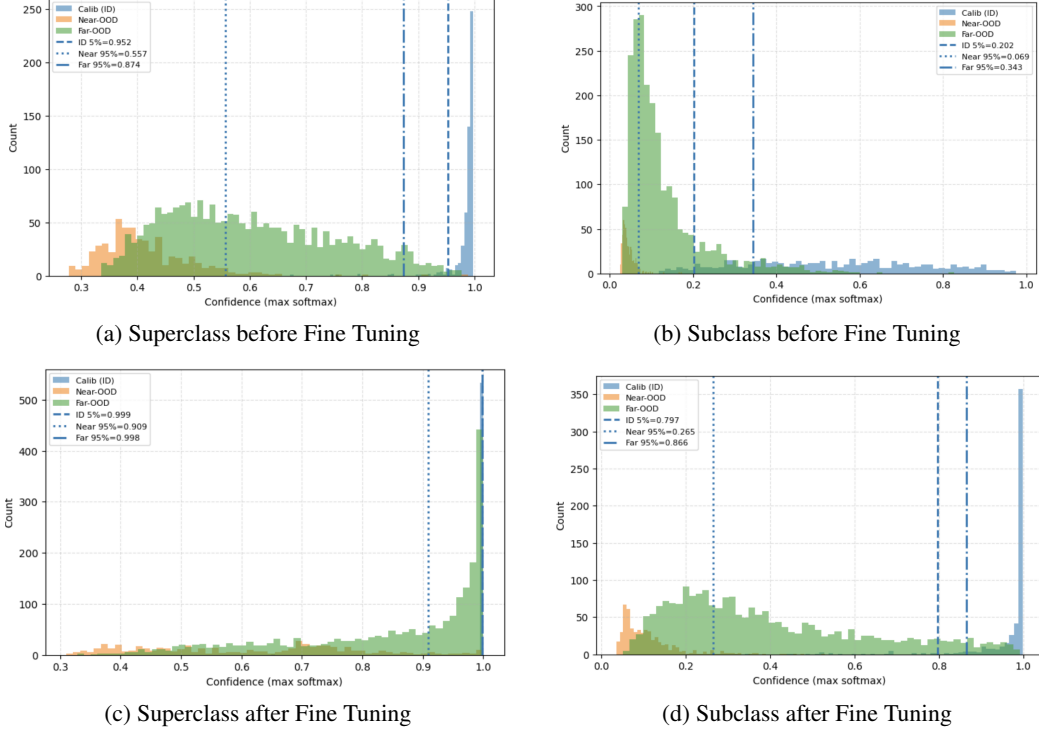


Figure 2: Confidence-Based OSR Calibration with CLIP

Let $f(x) \in \mathbb{R}^d$ denote the normalized embedding of an input sample x , and let $\{\mu_c\}_{c=1}^C$ denote class prototypes computed as the mean embedding of training samples from class c , also ℓ_2 -normalized. We define the cosine distance between x and class c as:

$$d_{\cos}(x, c) = 1 - \frac{f(x)^\top \mu_c}{\|f(x)\| \|\mu_c\|}.$$

The prototype-based open-set decision is given by:

$$\hat{y}(x) = \begin{cases} \text{novel}, & \text{if } \min_c d_{\cos}(x, c) > \tau_{\text{proto}}, \\ \arg \min_c d_{\cos}(x, c), & \text{otherwise,} \end{cases}$$

where τ_{proto} is a distance threshold calibrated using in-distribution and out-of-distribution validation data.

5.4 Ensemble OSR

With hopefully two good OSR detectors, we also decided to incorporate an ensemble method to improve recognition accuracy. We first decided to implement a metric-based fusion where decisions are weighted manually, then we decided to move on to simple AND voting.

We initially explored a metric-level fusion strategy that combines confidence-based and prototype-based scores into a single scalar decision function. Let $s_{\text{conf}}(x)$ denote the maximum softmax confidence for input x , and let $s_{\text{proto}}(x) = -\min_c d_{\cos}(x, c)$ denote a prototype similarity score derived from cosine distance. The fused score is defined as:

$$s_{\text{fused}}(x) = w_{\text{conf}} s_{\text{conf}}(x) + w_{\text{proto}} s_{\text{proto}}(x),$$

where w_{conf} and w_{proto} are manually selected scalar weights. A sample is rejected as novel if $s_{\text{fused}}(x) < \tau_{\text{fused}}$.

In practice, this approach was unstable due to incompatible score scales and highly overlapping distributions between in-distribution and out-of-distribution samples. Performance was sensitive to

weight selection, and no principled calibration strategy emerged. Consequently, metric-level fusion was abandoned in favor of decision-level fusion.

We ultimately adopted a decision-level fusion strategy based on logical conjunction, where if and only if both confidence-osr and prototype-osr votes novel is the sample classified as novel.

6 Experiments and Results

6.1 Confidence-Based OSR Performance

Across all architectures, confidence-based OSR achieves strong and consistent performance at both the superclass and subclass levels. Seen-class accuracy remains high, while unseen-class detection is reliable for both near-OOD and far-OOD samples, with stable behavior across backbones.

Figure 2 illustrates this separation. In-distribution samples concentrate at high confidence values, whereas OOD samples exhibit significantly lower confidence. The separation is especially pronounced for far-OOD data, validating the threshold calibration strategy. Near-OOD samples show partial overlap with ID confidence scores, reflecting their semantic proximity, yet are still detected with high recall.

Confidence-based OSR performs particularly well at the superclass level, where coarse semantic structure leads to clearer confidence separation. Although subclass-level OSR is more challenging due to fine-grained class overlap, confidence thresholding maintains competitive seen accuracy while rejecting a substantial fraction of unseen subclasses. Overall, these results establish confidence-based OSR as a strong and reliable baseline.

6.2 Prototype-Based OSR Performance

Prototype-based OSR exhibits a pronounced failure mode in our experiments. While it achieves near-perfect detection of unseen samples, it simultaneously suffers from a severe collapse in seen-class accuracy, rejecting the majority of in-distribution samples as novel at both the superclass and subclass levels.

Relaxing the prototype distance threshold across a wide range of values did not yield a meaningful recovery in seen-class performance. Even at substantially lower thresholds (e.g., 0.7/0.3), corresponding to a large relaxation in cosine space, most in-distribution samples remained distant from their class prototypes, indicating that the failure is not due to overly conservative thresholding.

This behavior reflects a fundamental mismatch between the assumptions of cosine prototype-based OSR and the learned embedding space. Prototype averaging collapses complex, multi-modal class structure, yielding unreliable distance-based novelty signals. Notably, this failure persists even for CLIP, indicating that its performance gains stem from improved discriminative representations rather than favorable metric geometry. Consequently, prototype-based cosine OSR is ineffective in this setting.

6.3 Ensemble OSR Performance

Despite the poor standalone performance of prototype-based OSR, we explored ensemble strategies as an exploratory analysis to determine whether combining confidence-based uncertainty with prototype distance could improve open-set recognition.

We first evaluated a metric-level fusion approach that combined confidence scores and prototype similarity using manually selected weights. Learning fusion weights from accuracy was considered but rejected, as the instability of prototype-based OSR would cause convergence to a trivial confidence-only solution. Even with manual weighting, metric-level fusion proved unstable due to incompatible score scales and overlapping ID and OOD distributions, leading to high sensitivity to weight selection and its eventual abandonment.

We then evaluated a decision-level ensemble based on logical conjunction, where a sample is classified as novel only if both confidence-based and prototype-based detectors indicate novelty. While this strategy prevented the catastrophic false positives of prototype-based OSR, it did not improve performance over confidence-only OSR. In practice, the prototype detector frequently rejected in-

Table 1: Open-set recognition performance for CLIP ViT-B/32 using different OSR methods.

OSR Method	Level	Seen (%)	Unseen (%)	Overall (%)
CLIP/B-32 (baseline)	Superclass	99.14	32.91	80.42
	Subclass	60.75	0.24	13.10
Confidence-based	Superclass	100.0	89.98	92.55
	Subclass	93.22	97.84	96.66
Prototype (Cosine)	Superclass	0.0	100.0	74.33
	Subclass	0.0	99.67	74.33
Fused (AND Voting)	Superclass	100.0	89.98	92.55
	Subclass	88.35	91.70	90.84

distribution samples, causing the ensemble to behave as an overly strict confidence threshold and reducing seen-class accuracy without meaningful gains in unseen detection.

Overall, ensemble strategies failed to improve upon the confidence-based baseline, indicating that fusion cannot compensate for a fundamentally misaligned novelty signal.

Table 1 reports open-set recognition performance for the CLIP ViT-B/32 backbone using confidence-based, prototype-based, and fused OSR methods, along with baseline results. Confidence-based OSR achieves the best overall tradeoff, maintaining high seen-class accuracy while reliably detecting unseen samples at both the superclass and subclass levels. In contrast, prototype-based cosine distance OSR exhibits a pronounced failure mode: although it achieves near-perfect unseen detection, it catastrophically rejects in-distribution samples, resulting in zero seen-class accuracy. The fused AND-voting approach does not improve performance over confidence-based OSR and slightly degrades subclass accuracy, indicating that ensembling cannot compensate for fundamentally misaligned novelty criteria. These results establish confidence-based OSR as the most effective and robust method in this setting.

6.4 Exploratory Alternatives

In addition to the OSR methods evaluated in this work, we also carefully considered several alternative approaches. These methods were not implemented due to time and scope constraints, rather than a lack of relevance, and were selected specifically to address the failure modes observed in prototype-based OSR.

One prominent alternative is Mahalanobis distance-based OSR, which models class-conditional feature distributions using both means and covariance matrices. Unlike cosine prototype methods, this approach can capture anisotropic and overlapping class structure and has demonstrated strong performance on the leaderboard when paired with ViT-based representations.

We also considered energy-based OSR methods, which derive novelty scores directly from classifier logits and can improve separation between in-distribution and out-of-distribution samples without relying on explicit distance metrics. EVT-based approaches such as OpenMax were additionally examined, as they explicitly model tail behavior of activation distributions to better handle uncertainty near class boundaries.

Finally, density-based approaches like k-nearest neighbor methods were considered for their ability to model complex embedding distributions. While these methods introduce additional computational and modeling complexity, they represent promising directions for refining and optimizing towards the OSR task.

Overall, these exploratory alternatives highlight that open-set recognition performance is closely tied to the geometric properties of the learned embedding space. Different representation landscapes favor different OSR mechanisms, and no single approach is universally optimal. Methods that model uncertainty or feature distributions behave differently depending on how class structure is organized in the embedding space, underscoring the importance of aligning OSR strategy with representation characteristics.

7 Conclusion

Throughout this project, we investigated transfer learning and open-set recognition in a hierarchical image classification setting under limited labeled data. We evaluated multiple pretrained backbones, including EfficientNet, DenseNet, ResNet, and CLIP, and found that CLIP consistently provided the strongest performance. In addition, confidence-based open-set recognition proved to be a robust and reliable mechanism for detecting novel classes, while prototype-based cosine distance methods failed by rejecting a large fraction of in-distribution samples, even when strong vision–language representations were used. This behavior suggests that averaging image embeddings into class prototypes collapses complex class structure, leading to misleading distance-based novelty signals in fine-grained tasks. In contrast, softmax confidence more effectively captures model uncertainty under distribution shift, enabling threshold-based rejection without sacrificing in-distribution classification accuracy. Attempts to combine confidence and prototype signals through fusion strategies did not improve performance and, in some cases, amplified failure modes, indicating that ensembling cannot compensate for fundamentally mismatched novelty criteria. Overall, our results show that CLIP, when paired with confidence-based open-set recognition, provides an effective and practical model for fine-grained classification and novel-class detection across both superclass and subclass levels.

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